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C23-M–105

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – FIRST YEAR EXAMINATION

ENGINEERING MECHANICS

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State *(a)* triangle law of forces and *(b)* polygon law of forces.

2. Two forces 8 N and 6 N act simultaneously at a point. Find the magnitude of resultant force, if the angle between them is 90°.

3. State any three laws of friction.

4. At limiting conditions, the angle of friction is 20°, find *(a)* the coefficient of friction and *(b)* angle of repose.

5. Find the moment of inertia of a rectangular lamina of 30 mm wide and 80 mm deep about Centroidal K-axis (I_{yy}).

6. Define the terms *(a)* centroid and *(b)* center of gravity.

7. A car starts from rest and attains a velocity of 20 m/s in 20 seconds. Find its *(a)* acceleration and *(b)* displacement during this time.

8. State the law of conservation of momentum and write mathematical expression for it.

9. Define the terms *(a)* mechanical advantage and *(b)* velocity ratio in case of simple machines.

10. Define the terms *(a)* kinematic chain and *(b)* mechanism.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. The following forces act at a point, Find the magnitude and direction of a resultant of force. 10

(a) A force of 25 N inclined at 35° towards north of east,

(b) A force of 22 N towards north,

(c) A force of 30 N towards north-west,

(d) A force of 20 N inclined at 35° towards south to west.

12. A body resting on a rough horizontal plane required a pull of 90 N inclined at 25° to the plane just to move it. It was also found that a push of 110 N inclined at 25° to the plane just moved the body. Determine the weight of the body and the coefficient of friction. 10

13. Find the moment of inertia of T-section having top flange 60 mm × 20 mm and web 70 mm × 20 mm, about its centroidal axes. 10

14. *(a)* A L-section is specified as 100 mm × 80 mm × 10 mm, find its centroid. 5

(b) The resultant of two concurrent forces is 12 N. If the forces are equal and makes 120° with each other, find their magnitude and the angle that resultant makes. 5

15. A particle moving in a straight line with uniform acceleration travels 7 m in the fourth second of its motion and 10 m in the eighth second. Find its initial velocity, acceleration and the distance travelled in the tenth second. 10

16. A wheel rotating at 30 rev/min is uniformly accelerated for 1·5 minutes during which time it takes 75 revolutions. What is the angular velocity of the wheel at the end of this interval and the further interval required for the wheel to reach an angular velocity of 90 rev/min. 10

17 Define Inversions of a mechanism and explain beam engine with a neat sketch. 10

18. In a lifting machine, an effort of 100 N lifts a load of 1500 N and an effort of 170 N lifts a load of 2700 N.

(a) Establish the law of machine;

(b) Calculate the effort required to lift a load of 4000 N.

(c) Find the load that can be lifted using an effort of 200 N.

(d) Find the maximum efficiency of the machine at velocity ratio of 75. 10

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